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TONE ALTERNATION IN KIRIKA (NKỌRỌỌ) ASSOCIATIVE CONSTRUCTION

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Associative construction in Kirika (formerly known as Nkọrọọ) is formed through the combination of two nouns accompanied by tonal alternation. Previous research proposed that this construction exhibits a fixed high-low tonal melody, except when it begins with a low tone, in which case the low tone is predicted to extend over the entire construction. However, this hypothesis could not be fully tested due to limited data. Using newly collected datasets, we evaluate this prediction. Our findings show that tonal alternation is restricted to the second noun. Moreover, when the construction begins with a low tone, the low tone does not spread to the end of the construction. Specifically, the second noun surfaces with a high tone on its initial mora and low tones on all subsequent morae. If the first noun ends in a low tone, the entire second noun surfaces with low tone. On the other hand, if the first noun ends in a high tone and the second noun is monomoraic, the latter noun then surfaces with a high tone. We present a formal analysis of the alternation. Thus, this work contributes to the description and analysis of Kirika.

Keywords: associative construction, grammatical tone, Ijoid, tone, syllable

1. Introduction

This work focuses on the pattern of tonal alternation in the associative construction in Kirika (ISO 639-3: nkx; Glottolog: nkor1239), which previous studies refer to as Nkọrọọ. Considering that the people refer to themselves as Nkọrọọ and their language as Kirika [kìrìkà] (Obikudo, 2008), we will refer to the language as such. Kirika is a tone language, which means pitch contrast distinguishes the meaning of one word from another. The language contrasts three tones, namely high (H), low (L), and downstep high (ꜜH). Like other closely related Ijoid languages, Kirika derives associative constructions by compounding two nouns, in which the first noun (N1) acts as the possessor while the second noun (N2) acts as the possessum. Akinlabi et al. (2009) describe and analyze various patterns of nominal constructions in Kirika, including the associative construction. The descriptive generalization of Akinlabi et al. (2009) is that the composite of nouns in an associative construction surfaces with a HL tonal melody but the entire construction surfaces with a L tone when the initial tone of the first noun is a L tone. Their descriptive generalizations are captured in Table 1i. Based on the pattern in their work, nouns undergo a tonal alternation when they occur as the second noun in an associative construction but do not undergo any tonal alternation as the first noun. Specifically, when the final TBU of the first noun bears a H tone, the initial TBU of the second noun bears a H tone and every non-final TBU bears a L tone. However, when the final TBU of the first noun bears a L tone, every TBU of the second noun surfaces with a L tone. In cases where the first tone of the first noun is L, every other tone-bearing unit of the construction bears a L tone.

Akinlabi and his colleagues did not find an associative construction in which the first noun has a LH tone melody in their data set, as this pattern is rare in the language. Consequently, they were unable to test the hypothesis that once the initial TBU of the associative construction begins with a L tone, "it continues to the end of the associative construction" (Akinlabi et al. 2009:11). They also did not find associative constructions where the second noun is monosyllabic. By presenting new data, the present study tests the hypothesis that once an associative construction begins with a L tone, every subsequent TBU in the construction bears a L tone. The new dataset covers two main patterns shown in Table 1ii: multimoraic first noun in (a) and monomoraic second noun in (b), accounting for the data that previous researchers had no access to. Table 1ii also contains the predicted patterns based on the hypothesis of Akinlabi et al. (2009).

Table 1: Tone patterns: Observations, new datasets, and predictions

i.	Observation					ii.	New datasets			Prediction	
	N1	N2	→	N1's	N2		N1	N2	→	N1's	N2
a.	HH	HH		HH	HL	a.	LH	HH		LL	LL
	HH	LL		HH	HL		LH	LL		LL	LL
	HH	HL		HH	HL		LH	HL		LL	LL
	HH	H$^{\downarrow}$H		HH	HL		LH	LH		LL	LL
	HH	LH		HH	HL		HLH	HH		HL	LL
b.	HL	HH		HL	LL		HLH	LL		HLL	LL
	HL	LL		HL	LL		HLH	LH		HLL	LL
	HL	LH		HL	LL		HLH	LH		HLL	LL
c.	LL	HH		LL	LL		LHL	HH		LL	LL
	LL	HL		LL	LL		LHL	LL		LL	LL
	LL	LL		LL	LL		LHL	HL		LL	LL
							LHL	LH		LL	LL
						b.	HL	H		H L	L
							LH…	H		LL	L
							LH…	L		LL	L
							HL…	L		LL	L

We will show that nouns do not undergo any tonal alternation when they occur as the first noun of associative construction, regardless of whether they bear LH, LHL or any other tone melody in isolation. Thus, the prediction does not hold that once a L tone begins an associative construction, it continues to the end of the compound (Akinlabi et al. 2009:11). Instead, our observation is that the final tone of the first noun determines whether the second noun surfaces with a H, L or a sequene of HL tones. We account for this alternation based on Rolle (2018)'s framework of grammatical tones, which are tonal operations that are restricted to morphosyntactic contexts.

Before turning to the dataset that forms the basis of our exploration, we present relevant background on the ethnography as well as the syllable structure and tone inventory of Kirika in Section 2. In Section 3, we present our data on associative construction. Our analysis of the patterns is presented in Section 4. The summary and conclusion of the discussion are presented in Section 5.

2. Background

Kirika (ISO Code: nkx; Glottolog: nkor1239) is an Eastern Ịjọid language of the Ijoid family within the Niger-Congo phylum. The language is spoken natively in Nkoro town of Opobo/Nkoro Local Government Area (LGA) of Rivers State, Nigeria. The map in Figure 1 shows their geographical location and the surrounding towns. Based on UNESCO Language Vitality and Endangerment factors, Kirika may be assessed as stable but threatened (Brenzinger and de Graaf, 2006). The language is spoken in most contexts by all generations, yet multilingualism in native and administrative languages such as Nigerian Pidgin and English has taken over most communicative domains. The language is generally understudied, but there is unpublished grammar, a few grammatical descriptions, word lists and texts useful for linguistic research (Obikudo, 2008, 2013, 2023). Audio and video recordings of Kirika also exist in varying quality, with and without annotation. The present work is based on data from Endangered Language Archive (Akinlabi et al., 2013) and Akinlabi et al. (2009). The data from the archive were collected during research trips to Nkoro town between 2008 and 2012. To supplement the existing data sets, the second author collected additional data from Dr. Bapakaye Peter in 2024.

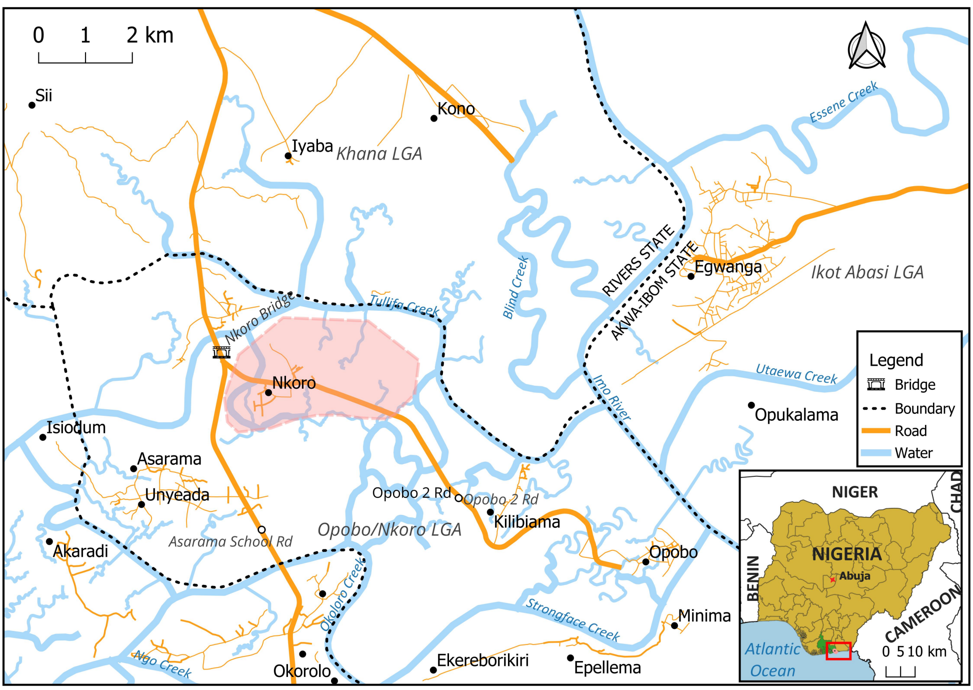

Figure 1: The native community of Kirika speakers in Nkoro town of Opobo/Nkoro LGA

2.1. Syllables and tone contrast

A syllable in Kirika is constructed either of a syllabic nasal (N), a vowel (V) or a composite of an onset consonant and a vowel (CV), as shown in (1). The vowels contrast in nasality, as seen in the second example in (1c). Since long vowels in Kirika are derived, studies suggest they are not contrastive (Williamson, 1978). As a result, we have represented long vowels as a sequence of two identical vowels, such as [tàà] 'wife' in (1). The language also permits syllables with two different vowels, such as [dùɔ̀] 'farm' in in (1d). Kirika does not have consonant clusters, but there are voiced prenasalized plosives (NC) that occur word-medially, as shown in (1e).

(1) Syllables in Kirika

a.	ńdó	'breast'	ŋḿgbà	'oyster'
b.	ìɓì	'be good'	àgbàká	'shoe'
c.	ɓó	'come'	ɓú	'body', 'self'
d.	dùɔ̀	'farm'	tàà	'wife'
	fìì	'death'	fĩ̀ĩ̀	'fly'
e.	mìndì	'water'	nìŋgì	'mother'
	nàmbùlò	'cow'		

Based on the distribution of vowels and consonants, we propose the syllable template CV(V)/N for Kirika. A moraic presentation of the syllable structure in Kirika is presented in Figure 2. Within the moraic theory, a short vowel and a syllabic nasal project one mora each, which is a timing unit within a syllable (Hyman, 1985; McCarthy, 1986; Hayes, 1989). A sequence of two vowels projects two morae, as seen in (1c). Given that the vowels in the language do not contrast in length, a long vowel is structurally a sequence of two monomoraic vowels. Just as Akinlabi et al. (2009:11), we found only a few monosyllabic nouns in the language. All the monosyllabic syllables we found are either monomoraic or bimoraic, as seen in (1c-d).

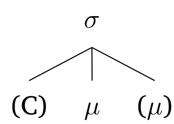

Figure 2: Moraic structure of Kirika syllable

Like all documented Nigerian languages, except Fulfulde, Kirika is a tone language, contrasting H, L and ꜜH tones (Obikudo, 2008). H and L tones are phonetically marked with acute (ó) and grave (ò) diacritics respectively. The downstep H is marked with a downward arrow [ꜜ] prefixed to an acute diacritic (ꜜó). The tone contrast in Kirika is illustrated in Table 2. The examples show that the H and L tones can occur in all environments, but the downstep H occurs only after a H tone.

Table 2: Tones in Kirika

			Second tone		
			H	L	DH
First tone	H	ɓú	ɔ́lɔ́	kírì	ɔ́ꜜlɔ́
		'body'	'phlegm'	'land'	'land'
	L	sɛ̀	tɔ̀kú	kìrì	
		'clan'	'child'	'slice'	
	ꜜH				

We consider the mora as the tone-bearing unit (TBU) in Kirika, under the assumption that there is a one-to-one relation between a tone and a TBU. The first evidence in support of this proposal is that the only syllables with a sequence of two tones are bimoraic, such as [fɔ̋ɔ̏] 'dust, ashes' in (2a) and the first syllable of [pɔ̀ɔ́sì] 'cat' in (2b). Another evidence is that the bimoraic syllables exhibit all possible tonal combinations. Under this account, two distinct level tones on a syllable constitute a contour tone.

(2) Tones of bimoraic syllable

a.	da̋a̋	'arrow'	fɔ̋ɔ̏	'dust, ashes'
b.	bɔ̀ɔ̀kɔ̀	'chicken'	pɔ̀ɔ́sì	'cat'
c.	póꜜó	'burn, spark'	m̀bɔ̀ḿ	'spitting cobra'

In the next section, we present the main focus of this work, which is the pattern of tonal alternation in associative constructions in the language. It should be mentioned that all the data sets presented in this work are in phonetic transcription.

3. Data presentation

3.1. Associative construction

An associative construction in Kirika is formed with a sequence of two nouns. The first noun (N1) functions as a possessor, and the second noun (N2) functions as a possessum. Regardless of their tones in isolation, the first noun does not undergo any tonal alternation, but the second noun shows three patterns of tonal alternation. Using the examples in (3-6), we illustrate the tonal alternation. The first noun in all the examples contain all tonal possibilities in the language. The examples are grouped based on the tone and syllable structures of the second nouns in isolation. In (3), the second nouns bear a H tone on every TBU in isolation, whereas in (4), the second nouns bear a L tone on every TBU in isolation.

(3) Multisyllabic N2 with H on all TBUs in isolation

a.	ɓú	'body'	díꜜmɛ́	'hair'	ɓú dímɛ̀	'hair on body'
	ɓéɓé	'mouth'	pákárá	'piece'	ɓéɓé pákàrà	'lip'
	ɓú	'body'	ápárá	'skin'	ɓú ápàrà	'body skin'
	òtìɔ́	'path'	ɓéɓé	'mouth'	òtìɔ́ ɓéɓè	'doorway'
	kɛ́ɓɛ̀lɛ́	'clay pot'	fítúrú	'bread'	kɛ́ɓɛ̀lɛ́ fítùrù	'claypot's bread'
	àkárá	'setting net'	dʒírí	'book'	àkárá dʒírì	'setting net's book'
b.	òbìrì	'dog'	téɓé	'head'	òbìrì tèɓè	'dog's head'
	òbìrì	'dog'	díꜜmɛ́	'hair'	òbìrì dìmɛ̀	'dog's hair'
	òbìrì	'dog'	pákárá	'piece'	òbìrì pàkàrà	'dog's piece'
	kírì	'land'	námá	'animal'	kírì nàmà	'land animal'
	kpàkóò	'baboon'	pákárá	'piece'	kpàkóò pàkàrà	'baboon's piece'

(4) Multisyllabic N2 with only L tone on all TBUs

a.	ámá	'town'	kàsì	'chair'	ámá kásì	'town's chair'
	kíní	'person'	òkpò	'back'	kíní ókpò	'person's back'
	kíní	'person'	àbànà	'pit'	kíní ábànà	'grave'
	tɔ̀kú	'child'	àkpà	'bag'	tɔ̀kú ákpà	'womb'
	láꜜkʷɔ́	'bile, gall'	àkpà	'bag'	láꜜkʷɔ́ ákpà	'gall bladder'
	àkárá	'setting net'	fìrì	'work'	àkárá fírì	'setting net's work'
b.	wárì	'house'	àkpànì	'side'	wárì àkpànì	'wall'
	pɔ̀ɔ́sì	'cat'	fìrì	'work'	pɔ̀ɔ́sì fìrì	'cat's work'
	ɓèrì	'ear'	pùlò	'oil'	ɓèrì pùlò	'earwax'
	dʒìnà	'urine'	àkpà	'bag'	dʒìnà àkpà	'bladder'

All the second nouns in (3–4) are at least bisyllabic. As the possessum of the associative constructions in (3a–4a), the second nouns bear a H tone on their initial TBU and a L tone on every non-initial TBUs when the last tone of the first noun is H. However, when the last tone of the first noun is L, the second noun bears a L tone on all TBUs, as shown in (3b–4b). This tonal alternation applies to nouns with at least two syllables, regardless of their lexical tone melody. Despite having dissimilar lexical tones, multisyllabic second nouns in surface with a HL melody when the final tone of the first noun is H, as seen in (5a), and with a L tone on every TBU when the final tone of the first noun is L, as seen in (5b).

(5) Multisyllabic N2 with different tones

a.	íré	‘female’	tɔ̀kú	‘child’	íré tɔ́kù	‘girl’
	námá	‘animal’	àgbàká	‘shoe’	námá ágbàkà	‘animal hoof’
	ówú	‘mask’	tɔ́rù	‘face’	ówú tɔ́rù	‘face mask’
	dʒírí	‘book’	wárì	‘house’	dʒírí wárì	‘school’
	ɔ́rɔ́rɔ́	‘female’	kpàkóò	‘baboon’	ɔ́rɔ́rɔ́ kpákòò	‘female baboon’
	námá	‘animal’	kɛ́ɓɛ̀lɛ́	‘clay pot’	námá kɛ́ɓɛ̀lɛ̀	‘animal’s claypot’
	ókpòkó	‘gizzard’	fúrò	‘inside’	ókpòkó fúrò	‘gizzard’s inside’
	kɛ́ɓɛ̀lɛ́	‘claypot’	kírì	‘ground’	kɛ́ɓɛ̀lɛ́ kírì	‘claypot’s ground’
	óɓórí	‘goat’	kɔ́kɔ̀lì	‘crab’	óɓórí kɔ́kɔ̀lì	‘goat’s crab’
b.	òbìrì	‘dog’	tɔ̀kú	‘child’	òbìrì tɔ̀kù	‘puppy’
	àbàtì	‘God’	àgbàká	‘shoe’	àbàtì àgbàkà	‘dog’s shoe’
	dùɔ̀	‘farm’	kírì	‘land’	dùɔ̀ kìrì	‘village’
	pìrì	‘wild’	kpàkóò	‘baboon’	pìrì kpàkòò	‘wild baboon’
	òbìrì	‘dog’	kɛ́ɓɛ̀lɛ́	‘clay pot’	òbìrì kɛ̀ɓɛ̀lɛ̀	‘dog’s claypot’
	pɔ̀ɔ́sì	‘cat’	fítúrú	‘bread’	pɔ̀ɔ́sì fìtùrù	‘cat’s bread’
	àtɔ́nì	‘money’	kɔ́kɔ̀lì	‘crab’	àtɔ́nì kɔ̀kɔ̀lì	‘money’s crab’

As discussed in Section 2.1, monosyllabic words are either bimoraic or monomoraic. When serving as the second noun in an associative construction, monosyllabic words with two moras pattern like words with two or more syllables. However, monomoraic words follow a different pattern. In (6), we illustrate the effect of syllable-type on the tonal alternation of the associative construction in Kirika.

(6) Monosyllabic N2 in associative construction

a.	òbìrì	‘dog’	tĩ̋	‘stick’	òbìrì tĩ̏	‘dog’s stick’
	òbìrì	‘animal’	tàà	‘wife’	òbìrì tàà	‘animal’s wife’
	wárì	‘house’	fìì	‘death’	wárì fìì	‘dying at home’
b.	téɓé	‘head’	tĩ̋	‘stick’	tébé tĩ̂	‘head’s stick’
	tɔ̀kú	‘child’	tàà	‘wife’	tɔ̀kú táà	‘daughter-in-law’
	tɔ̀kú	‘child’	ɓíɔ́	‘leg’	tɔ̀kú ɓíɔ̀	‘child’s leg’
c.	òbìrì	‘dog’	ɓú	‘body’	òbìrì ɓù	‘dog’s body’
	òbìrì	‘dog’	sɛ̀	‘clan’	òbìrì sɛ̀	‘dog’s clan’
	kɔ́kɔ̀lì	‘crab’	jé	‘thing’	kɔ́kɔ̀lì jè	‘crab’s thing’
d.	íré	‘female’	ɓú	‘body’	íré ɓú	‘female body’
	kíní	‘person’	sɛ̀	‘clan’	kíní sɛ́	‘person’s clan’
	tɔ̀kú	‘child’	jé	‘thing’	tɔ̀kú jé	‘child’s thing’

Monosyllabic words with two morae bear a L tone on every TBU when the final tone of the first noun is L, as seen in (6a). However, when the final tone of the first noun is high, monosyllabic words with two morae bear a sequence of high-low tones, as shown in (6b).

As illustrated in (6c), monomoraic words follow the same pattern as other words when the final tone of the first noun is L, bearing a L tone. However, when the final tone of the first noun is H, the second noun bears a H tone regardless of its tone in isolation, as seen in (6d). In other words, the distinction between monomoraic and multimoraic second nouns (including those with at least two syllables) is tied to the final H tone of the first noun. In the next section, we present the generalizations of the alternation.

3.2. Descriptive generalizations

The discussion shows that the tone melodies of the second nouns of the associative construction are either L, HL or H. Before presenting an analysis of the nominal modification, the descriptive generalizations of the tonal alternation are presented in this section.

(7) Tonal alternation in associative construction: L, H or H-L tones of the entire second noun

a. Regardless of its syllable type, the second noun bears a L tone on every TBU when the final tone of the first noun is L.

b. Polymoraic second nouns bear a H tone on their initial TBU and a L tone on every other TBU when the final tone of the first noun is H.

c. Monomoraic second nouns bear a H tone when the final tone of the first noun is H.

d. In all instances, the final tone of the first noun and the initial tone of the second noun are identical.

As mentioned in Section 1, the main goal of this work is to test the hypothesis of Akinlabi et al. (2013:11) that once the initial TBU of the associative construction begins with a L tone, "it continues to the end of the compound". Given that they had no access to first nouns with LH tones, Akinlabi et al. (2009) were unable to test the hypothesis. In Table 3, the observed patterns of the present work and the data gaps of the previous study are illustrated.

Table 3: Tonal concatenation and alternations of the possessum in the associative construction

		Second Noun (N2)							
		H	L	HH	HꜜH	LHL	LH	HLH	LHL
First Noun (N1)	H	H-H	H-H	H-HL	H-HL	H-HL	H-HL	H-HLL	H-HLL
	L	L-L	L-L	L-LL	L-LL	L-LL	L-LL	L-LLL	L-LLL
	HH	HH-H	HH-H	HH-HL	HH-HL	HH-HL	HH-HL	HH-HILL	HH-HLL
	HꜜH	HꜜH-H	HꜜH-H	HꜜH-HL	HꜜH-HL	HꜜH-HL	HꜜH-HL	HꜜH-HLL	HꜜH-HLL
	LL	LL-L	LL-L	LL-LL	LL-LL	LL-LL	LL-LL	LL-LLL	LL-LLL
	HL	HL-L	HL-L	HL-LL	HL-LL	HL-LL	HL-LL	HL-LLL	HL-LLL
	LH	LH-H	LH-H	LH-HL	LH-HL	LH-HL	LH-HL	LH-HLL	LH-HLL
	HLH	HLH-H	HLH-H	HLH-HL	HLH-HL	HLH-HL	HLH-HL	HLH-HLL	HLH-HLL
	LHL	LHL-L	LHL-L	LHL-LL	LHL-LL	LHL-LL	LHL-LL	LHL-LLL	LHL-LLL

The shaded cells contain data that Akinlabi et al. (2009) did not have access to. The observed patterns in this work are not consistent with their prediction. Instead, our findings show that the tone associated with the final TBU of the first noun, along with the syllable structure of the second noun, determines whether the tonal alternation of the second noun involves a L tone, a H tone, or a sequence of HL tones. In the next section, we address each of these generalizations to account for the observed tonal alternations.

4. Analysis

4.1. Grammatical tone

The tonal alternations of the associative constructions in Kirika can be considered a pattern of grammatical tone, which is a tonal operation that is restricted to a morphosyntactic context (Rolle, 2018). This pattern of grammatical tone is found in the associative constructions of many Benue-Congo, Chadic and Nilo-Saharan languages (Cahill, 2000; Hellwig, 2011; Schrock, 2014; Akinbo and Bulkaam, 2024; Griffin et al., 2025). To capture the properties of grammatical tones across languages, Rolle (2018) proposes the following components:

(8) Components of grammatical tones (Rolle 2018:28–29):

a. Grammatical tune: the unique tone or set of tones that co-varies with the grammatical construction.

b. Trigger: the morphemes or constructions that license the tonal operation.

c. Target: the morpheme or morphemes that is the intended undergoer of a tonological operation.

d. Sponsor: the morpheme (or natural class of morphemes) that co-varies with the grammatical tune.

e. Host: the morpheme or morphemes on which the grammatical tune appears.

f. Valuation window: the portion of the target-host that is evaluated with respect to whether its TBUs are valued or unvalued; this can be coextensive with the target-host, or strictly a local subconstituent.

We account for the pattern of tonal alternation in Kirika using the components proposed by Rolle (2018). To explain the trigger of the grammatical tone, two possible approaches can be considered: (i) the entire associative construction as the trigger, or (ii) a segmentally null associative linker. As Rolle (2018:30) notes, these two accounts are potentially equivalent. In this work, we adopt the second approach, assuming that the trigger of the grammatical tone is a segmentally null associative linker whose sole exponent is a tone melody, as shown in (9). To determine the tonal exponent of the associative morpheme, we compare Kirika to other Ijoid languages. Previous studies suggest that associative constructions in Ijoid languages typically involve a sequence of HL tones (Williamson, 1978; Harry, 2004; Obikudo, 2010; Harry and Hyman, 2014). The tone melody is interpreted as a gradual shift of the languages to an accentual system. Based on this insight, Akinlabi et al. (2009) also propose that the grammatical tone of associative constructions in the language involves a sequence of HL tones. In line with Akinlabi et al. (2009), we also propose that the exponent of the associative linker in Kirika is a sequence of HL tones, as seen in (10). As with Akinlabi and his colleagues, we assume that the linear of the HL melody is prespecified in the input. As mentioned earlier, Akinlabi et al. (2013:11) suggest (i) that the second noun undergoes tonal alternation and (ii) that once an associative construction begins with a L tone, “it continues to the end” of the construction. This means that the L tone of the grammatical tone melody targets the first and second nouns of the associative construction. In line with Akinlabi et al. (2009), we also find that only the second noun undergoes tonal alternation in the construction. However, the new datasets show that the construction may not bear a L tone on every TBU whenever it starts with a L tone. As a result, we suggest that the target and host of the grammatical tone melody is the second noun, not the two nouns of the construction. The second noun is probably the target and host of the grammatical tone because the associative linker is a prefix just as many affixes in Kirika (Obikudo, 2013).

(9) Associative construction in Kirika
N1 Linker N2

(10) Exponent of the associative linker in Kirika
H L tones

We now turn to the valuation of the grammatical tune, beginning with second nouns that are at least bimoraic syllables. To account for the valuation of the grammatical tune, we must consider that the final tone of the first noun and the initial tone of the second noun are consistently identical. The tonal agreement can be attributed to a condition requiring the final tone of the first noun and the initial tone of the second noun to match. To ensure the initial tone of the second noun agrees with the final tone of the first noun, the H tone of the associative linker could have been realized on the initial and medial morae while the L tone is realized word-finally, as illustrated with unattested form in (11a). The optimal realization of the grammatical tone, as seen in (11b), suggests a left-to-right association of the grammatical tone. Also, as a result of the tonal agreement condition, the L tone of the grammatical tune is realized on every TBU of the second noun when the final TBU of the first noun bears a L tone, but the H tone of the grammatical tone melody is not realized in this instance. This means that the final tone of the first noun determines whether the H tone of the grammatical tune will be realized or not. In line with the proposal of Rolle (2018), the first noun can be considered the sponsor of the grammatical tune because it determines whether the second noun surfaces with a L tone or with a HL tonal sequence. This also means that the valuation windows of the H and L tones belonging to the grammatical tone melody are the initial TBU and every TBU of the second nouns, respectively.

(11) Realizing the HL melody in the context of a first noun with a final H tone

a. * H L
fu tu ru

b. H L
fu tu ru

An aspect of the grammatical tone operation, which is absent in Akinlabi et al. (2009), is the role of syllable type—particularly when the second noun is a monomoraic syllable. As shown in (6d), when the final tone of the first noun is H, monomoraic second nouns surface with a H tone rather than the HL melody observed in nouns with two or more morae. To account for the realization of only the H tone in this case, we must consider that the mora is the TBU in Kirika, given that the only syllables with a sequence of two tones are bimoraic. Thus, both H and L tones of the grammatical tune can only be realized simultaneously on a monosyllabic second noun with at least two morae. This leaves us with two options of realizing the H and L tones of the grammatical tune: (i) inserting a mora to lengthen the vowel of the monomoraic noun, or (ii) realizing only one tone of the grammatical tune. As seen in (6), the monomoraic second nouns do not undergo vowel lengthening but instead bear a H or L tone depending on the final tone of the first noun. This indicates that when the second noun is monomoraic, the language prefers to realize only one tone of the grammatical tune rather than epenthesizing a mora to host the second tone. Vowel lengthening is not the preferred option probably because the vowels of the language do not contrast in length. Like the bimoraic nouns, the tone of the grammatical tune that is realized on the monomoraic second noun depends on the final tone of the first noun. Specifically, the realized tone of the grammatical tune is obligatorily identical to the final tone of the first noun.

In general, a segmentally null morpheme with a sequence of HL tones as its exponent triggers the grammatical tone of the associative constructions in Kirika. The target of the grammatical tune is the second noun of the associative construction. In their realization, the H tone of the grammatical tune has the initial TBU of the second noun as its valuation window, while the L tone has all TBUs of the second noun as its valuation window. The fact that the H tone is realized on the initial TBU of the second noun and the L tone on its non-initial TBUs is attributed to a condition requiring the tone at the final edge of the first noun to agree with the tone at the initial edge of the second noun. Consequently, when the final tone of the first noun is L, the L tone of the grammatical tune is realized on the entire second noun. Since the language enforces a one-to-one correspondence between a tone and a TBU, which in this case is a mora, only one tone of the grammatical tune is realized when the second noun is monomoraic. The tone that surfaces in such cases is identical to the final tone of the first noun, also due to the tonal agreement between the final and initial TBUs of the first and second nouns, respectively. In the next section, we present a formal account of the grammatical tune.

4.2. A formal account of the grammatical tone

We formalize our account of the grammatical tone in this section within the framework of Optimality Theory (Prince and Smolensky, 2004). Our goal is not to argue in favor of one specific formal account of grammatical tone but to simply illustrate how one may easily account for the alternation using classical OT constraints. We can formalize the realization and maximal extension of the grammatical tune using morpheme-specific anchor constraints as proposed by Finley (2009). We illustrate the operation of the constraint using a L-tone version of the constraint in (12). To capture the generalization that the target of the grammatical tone is the second noun (possessum), the constraints are defined such that their domain of operation is the possessum. The constraint L-Anchor-L_{link} in (12a) assigns no violations if the L tone of the linker is associated with the leftmost TBU of the possessum. Similarly, the constraint R-Anchor-L_{link} (12b) assigns no violations if the L tone of the linker is associated with the rightmost TBU of possessum. If the L tone of the linker is associated with a TBU of the possessum other than the leftmost one, the constraint L-Anchor-L_{link} assigns a violation to each TBU to the left of the realized tone. In a situation where the L tone of the linker is associated with a TBU of the possessum other than the rightmost TBU, the constraint R-Anchor-L_{link} assigns a violation to each TBU to the right of the realized tone. If the L tone of the linker is not realized on any TBU of the possessum, the constraint L-Anchor-L_{link} assigns a violation to each TBU of the possessum. The constraint R-Anchor-L_{link} also assigns a violation to each TBU of the possessum in the same situation. To categorically satisfy both L-Anchor-L_{link} and R-Anchor-L_{link}, the L tone of the linker must be realized on all TBUs of the possessum. We assume that the H tone version of the constraints also exists in the grammar.

(12) Morpheme-Feature Edge-Anchoring (Finley, 2009):

a. L-Anchor-L_{link}: the L tone of the associative linker must be in correspondence with the leftmost edge of the possessum.

b. R-Anchor-L_{link}: the L tone of the associative linker must be in correspondence with the rightmost edge of the possessum.

(13) Max-T: (Pulleyblank, 1997; Zoll, 2003):
'Every tone linked to a TBU in the input has an output correspondent'

Realizing the grammatical tone melody can result in the deletion or floating of lexical tones associated with the second noun. Given that we found no downstep on the initial H tone of the second noun in any context, we assume that the realization of the grammatical tones results in the deletion of lexical tones. The deletion of a tone will result in a violation of the constraint Max-T in (13), which requires every tone linked to a TBU in the input to have an output correspondent. The faithfulness constraint Max-T has to be ranked below the anchor constraints, given that the grammatical tone melody consistently replaces every lexical tone of the possessum.

In our analysis, the H tone of the grammatical tune has the initial TBU of the second noun as its valuation window, while the L tone has every TBU of the second noun as its valuation window. We can capture this property of the tones by ranking the constraints L-Anchor-H_{link} and R-Anchor-L_{link} above the constraint L-Anchor-L_{link}, which in turn must be ranked above R-Anchor-H_{link}. The ranking of R-Anchor-H_{link} below L-Anchor-L_{link} ensures that the L tone also applies to medial tone-bearing units. In (14), we illustrate our account of the tonal alternation when the first noun has a final H tone, and the second noun has at least two morae. We use bracket notation in place of autosegmental association lines, and numeral indexation represents the input-output mapping of tones.

(14) /ámá + fítúrú/→[ámá fítùrù] 'town's bread'

	$(ámá)_1$ + H_3L_{4LINK} + $(fítúrú)_2$	L-ANCH-H_{LINK}	R-ANCH-L_{LINK}	L-ANCH-L_{LINK}	R-ANCH-H_{LINK}	MAX-T
a.	$(ámá)_1$ $(fítúrú)_2$	*!**	*!**	***	***	
b.	$(ámá)_1$ $(fítú)_2(rù)_4$	*!**		**	***	*
c.	$(ámá)_1$ $(fí)_3(túrú)_2$		*!**	***	**	*
d. ☞	$(ámá)_1$ $(fí)_3(tùrù)_4$			*	**	*
e.	$(ámá)_1$ $(fí)_2(tùrù)_4$	*!**		*	***	*
f.	$(ámá)_1$ $(fítú)_3(rù)_4$			**!	*	*
g.	$(ámá)_1$ $(fìtùrù)_4$	*!**			***	*

For not realizing the L or H tone of the grammatical tune, the candidates in (14a-c) and (14e) are ruled out. The candidate in (14d) wins for a non-fatal violation of the constraints R-Anchor-H_{link} and L-Anchor-L_{link} because the H tone is realized on the initial TBU of the second noun and the L tone is realized on every non-final TBU of the second noun. Despite the surface similarity between the winner in (14d) and the loser in (14e), the anchor constraints are able to distinguish grammatical tones from lexical tones by assigning violation marks to the unparsed H tone of a grammatical tone melody.

(15) Agree-Edge[T] (Pulleyblank, 2002)
The tone at the right edge of the possessor must agree with the tone at the left edge of the possessum.

In (14), the winning candidate does not only involve realizing all tones of the grammatical tune but also involves a tonal agreement between the final tone of the first noun and the initial tone of the noun that undergoes the grammatical tone operation. In the previous section, the tonal agreement is considered the effect of a condition which requires the final tone of the first noun to agree with the initial tone of the first noun. We formalize the condition as the constraint Agree-Edge[T] in (15). By ranking the tonal agreement constraint above L-Anchor-H_{link} and R-Anchor-L_{link}, we can account for the tonal alternation when the final tone of the first noun is L. In (16), we illustrate our account of the alternation.

(16) /òbìrì + pákárá/→[òbìrì pàkàrà] 'dog's piece'

	(òbìrì)$_1$ + H_3L_{4link} + (pákárá)$_2$	Agree-Edge-T	L-Anch-H_{link}	R-Anch-L_{link}	L-Anch-L_{link}	R-Anch-H_{link}	Max-T
a.	(òbìrì)$_1$ (pákárá)$_2$	*!	***	***	***	***	
b.	(òbìrì)$_1$ (pá)$_3$(kárá)$_2$	*!		***	**	***	*
c.	(òbìrì)$_1$ (pá)$_3$(ká)$_2$(rà)$_4$	*!			**	**	
d.	(òbìrì)$_1$ (pá)$_3$(kàrà)$_4$	*!			*	**	*
e.	(òbìrì)$_1$ (páká)$_3$(rà)$_4$	*!			**	*	*
f. ☞	(òbìrì)$_1$ (pàkàrà)$_4$		***			***	**
g.	(òbìrì)$_1$ (pá)$_2$(kàrà)$_4$		***		*!	***	*

When the first noun has a final L tone, realizing all tones of the grammatical tune can create a situation where the final tone of the first noun and the initial tone of the second noun are not identical. This may result in a fatal violation of the condition Agree-Edge[T], as it is for the candidates in (16a–e). While the candidate in (16g) satisfies Agree-Edge[T], it is ruled out for failing to realize the L tone of the grammatical tune on the leftmost TBU. In contrast, by realizing the L tone of the grammatical tune on every TBU of the second noun, the winning candidate in (16f) satisfies both Agree-Edge[T] and R-ANCH-L_{link} by realizing the tone of the grammatical tune on the entire second noun.

(17) *Contour (Zoll 2003:236)
"Contours are licensed only on a long vowel."
(assign a violation to a mora with more than one tone).

So far, we have only shown cases where the second noun is at least bimoraic. We now turn to cases where the second noun is monomoraic. As shown earlier, the HL tones of the grammatical tunes are fully realized on the second noun when the final tone of the first noun is H. However, only one tone of the grammatical tone is realized on the monomoraic nouns, due to the mora being the TBU in the language and a one-to-one relation between the TBU and tone. We formulate the tone-TBU relation as the constraint *Contour in (17), which assigns violation to a mora with two or more tones. Given that there are no monomoraic syllables with a contour tone, the constraint *Contour is ranked above all the constraints but not crucially ranked with respect to the constraint Agree-Edge[T]. In (18-19), we illustrate that the ranking of the constraints can account for the tonal alternation of monomoraic second nouns.

(18) /ámá + ɓú/→[ámá ɓú] 'town's body'

	(ámá)$_1$ + H$_3$L$_{4link}$ + (ɓú)$_2$	*Contour	Agree-Edge-T	L-Anch-H$_{link}$	R-Anch-L$_{link}$	L-Anch-L$_{link}$	R-Anch-H$_{link}$	Max-T
a.	(ámá)$_1$ (ɓú)$_2$			*!	*	*	*	
b.	(ámá)$_1$ (ɓù)$_4$		*!	*			*	*
c. ☞	(ámá)$_1$ (ɓú)$_3$				*	*		*
d.	(ámá)$_1$ (ɓû)$_{3,4}$	*!						*

(19) /pìrì + ɓú/→[pìrì ɓù] 'bush body'

	(pìrì)$_1$ + H$_3$L$_{4link}$ + (ɓú)$_2$	*Contour	Agree-Edge-T	L-Anch-H$_{link}$	R-Anch-L$_{link}$	L-Anch-L$_{link}$	R-Anch-H$_{link}$	Max-T
a.	(pìrì)$_1$ (ɓú)$_2$			*	*!	*	*	
b. ☞	(pìrì)$_1$ (ɓù)$_4$			*			*	*
c.	(pìrì)$_1$ (ɓú)$_3$		*!		*	*		*
d.	(pìrì)$_1$ (ɓû)$_{3,4}$	*!	*!					*
e.	(pìrì)$_1$ (ɓǔ)$_{4,3}$	*!						*
f.	(pìrìɓù)$_1$			***	***	***	***	*

The second nouns of the forms in (18) and (19) are monomoraic. For realizing the H and L tones of the grammatical tune on the monomoraic second nouns, the candidates in (18d) and (19d-e) incur a fatal violation of the constraint *Contour because the monomoraic second noun bears a sequence of two tones (indicated with a circumflex and caron diacritics for convenience). Although the candidates in (18b) and (19c) satisfy *Contour, they are ruled out for incurring a fatal violation of the constraint Agree-Edge[T], given that the initial tone of the monomoraic second noun is not identical to the final tone of the first noun. By realizing the tone of grammatical tune that is identical to the final tone of the first noun, the winning candidates in (18c) and (19b) satisfy *Contour and Agree-Edge[T]. In other words, the constraint ranking is able to select the second noun with the grammatical tone that is identical to the final tone of the first noun.

The realization of the HL grammatical tune in associative constructions such as [ámá fítùrù] 'town's bread' is consistent with left-to-right mapping (Goldsmith, 1976; Zoll, 2003). The directionality of tone mapping is traditionally formalized using markedness constraints such as Generalized Alignment constraints (Akinlabi, 1996; Zoll, 2003). However, Finley (2009) suggests that the realization of grammatical tone melody is better formalized using correspondence constraints because they are able to distinguish lexical and grammatical tones. In line with the suggestion, we have formalized our account of the grammatical tone using anchor constraints.

5. Summary and conclusion

We have described and analyzed the tonal alternation of associative constructions in Kirika based on existing and new datasets. According to Akinlabi et al. (2009), the composite of words in associative construction bears a HL melody but has a L tone on every TBU when the initial TBU of the first noun bears a L tone. Therefore, their prediction is that once the initial TBU of the associative construction begins with a L tone, "it continues to the end of the compound" (Akinlabi et al. 2009:11). However, they did not have access to enough data to test the hypothesis. We test this hypothesis by exploring all logically possible syllable and tonal combinations. We find that only the second noun of the associative construction undergoes tonal alternation. Regardless of their tone in isolation, the second noun bears a sequence of H and L tones, a L tone or a H tone, depending on the final tone of the first noun and the syllable type of the second noun. Specifically, when the final tone of the first noun is H, the second noun with at least two moras surfaces with a H tone on the first mora and a L tone on every other mora. In a situation where the final tone of the first noun is L, the second noun surfaces with a L tone on every TBU. However, monomoraic second nouns surface with a H tone when the final tone of the first noun is H. A crucial observation of the tonal alternation is that the initial tone of the second noun consistently agrees with the final tone of the first noun.

The mora is considered the TBU in Kirika. In our account of the tonal alternation, a segmentally null associative linker with a sequence of HL tones as its sole exponent is considered the trigger of the grammatical tune. The H tone of the grammatical tune is associated with the initial TBU of the second noun, and the valuation window of the L tone is every TBU of the second noun. The fact that the final and initial tones of the first and second nouns obligatorily agree is considered the effect of a tonal agreement condition. The realization of the H tone on the initial TBU of the second noun and the L tone on every non-final TBU of the second noun aligns with the tonal agreement condition. On the other hand, when the final tone of the first noun is L, the same tonal agreement causes the L tone of the grammatical tune to be realized on all TBUs of the second noun, leaving the H tone unrealized. That only the tone of the grammatical tune that agrees with the final tone of the first noun is realized on a monomoraic second noun also supports the effect of the agreement condition.

As mentioned earlier, a HL tonal melody, as well as a LH melody, is widely attested in the nominal modification of many Ijoid languages. Previous studies suggest that the tonal melody points to a gradual shift to an accentual system in the languages (Williamson, 1978; Harry, 2004; Akinlabi et al., 2009; Williamson, 2011; Harry and Hyman, 2014). Our account of the grammatical tone is also consistent with this hypothesis except that the tonal melody only applies to the second noun of the construction. A crucial contribution of our work is that the final tone of the first noun and the syllable type of the second noun jointly determine whether one or both tones of the grammatical tune are parsed in the construction. Therefore, our study highlights the importance of both syllable type and the final tone of the first noun to the realization of the grammatical tune of the construction in Kirika. This work also supports the proposal that the mora is the TBU in the language. The first evidence is that only bimoraic syllables bear two tones, and the second is that the HL grammatical tone melody requires at least bimoraic words for its realization.

The agreement between the final tone of the first noun and the initial tone of the second noun is similar to vowel harmony across morpheme boundaries (see Ritter and van der Hulst 2024). van der Hulst (2016:3) suggests that the agreement of vowel features across morpheme boundaries in vowel harmony can "serve as a marker of wordhood, since a switch [from one feature specification to another] can … indicate the division between words". By extension, the agreement between the final tone of the first noun and the initial tone of the second noun may indicate that the two nouns of the associative constructions constitute a word. Since Kirika, like other Ijoid languages, is understudied, our work contributes to the documentation and analysis of the language.

Abbreviations

H	High tone
L	Low tone
ꜜH	Downstep high tone
N1	First noun of an associative construction
N2	Second noun of an associative construction
TBU	Tone-bearing unit

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Author contributions

The authors were jointly responsible for the conceptualisation of the study, data collection, description and analysis. While the secondary data were compiled by the first author, the primary data were collected by the second author. They also co-drafted and revised the manuscript.

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